THE PHOTOELECTRIC EFFECT

This is the phenomenon that involves the emission of electrons from a metal when light or any other radiation of suitable wavelength (or frequency) strikes the surface.

The emitted electrons are called photoelectrons

Quantum Theory of Light

- The view that light propagates as a series of separate packets of energy (usually called quanta or photons) is directly opposed to the wave theory of light.
- Planck found that the quanta associated with a particular frequency v of light must all have the same quantum energy and this energy E is directly proportional to v, i.e.

$$E = hv$$

h is the Planck's constant given as 6.63×10^{-34} J s

- **While he had to assume that electromagnetic energy radiated by an object emerges intermittently, Planck did not doubt that it propagates through space as continuous electromagnetic waves.**
- **Einstein proposed that light not only is emitted a quantum at a time, but it also propagates as individual quanta, a more drastic break with classical physics.**
- **With regards to this hypothesis, the photoelectric effect can be readily explained.**

- **The empirical formula may be rewritten as:**

$$h\nu = K_{\max} + h\nu_0$$

$h\nu$ = **energy content of each quantum of the incident light**

$K_{\max}$ = **maximum photoelectron energy**

$h\nu_0$ = **minimum energy needed to dislodge an electron from the metal surface being illuminated**

- **Note**
- **There must be a minimum energy required by an electron in order to escape from a metal surface, or else electrons would pour out even when there is no presence of light.**

The energy characteristic of a particular surface is called its *work function*.

The photoelectric effect equation can be stated as follow:

$$h\nu = K_{max} + h\nu_0$$

Quantum energy = maximum electron energy + work function of surface

To express hν in electron volts, $1eV = 1.60 \times 10^{-19} J$

$1\overset{0}{A} = 10^{-10} m$ or 0.1nm

$$\nu = \frac{c}{\lambda}$$

c is the speed of light, ν is the frequency and λ is the wavelength

- **Einstein's proposal accounts for several unexplained features associated with this photoelectric effect .**
- **He proposed that;**
- **The maximum energy, $E_{\max}$ of emitted electrons is found to be**

$$E_{\max} = eV_0$$

where e is the charge of electron and V_0 is the stopping potential.

- **Experiment shows that:**

i. **When light strikes a metal surface, a current flows instantaneously, even for very weak light.**

ii. **At a fixed frequency, the strength of the current is directly proportional to the intensity of the light.**

iii. **The stopping potential V_0 and the maximum energy, E_{max} of the emitted electrons depends only on the frequency of the light and the type of material.**

iv. Each metal has a characteristics threshold frequency, v_0 such that;

$$eV_0 = h(v - v_0)$$

v. The constant *h* is found to be the same for all metals and turns out to be the same constant used by Planck in his blackbody derivation.

Hence, each of these experiment ideas can be explained by accepting that light is made up of particles.

- **Example**
- **Find the maximum kinetic energy in electron volt eV of the photoelectrons emitted when ultraviolet light of wavelength 3000 $\overset{0}{A}$ fall on a potassium surface (take work function of potassium as 2.50eV).**

- **Solution**
- $h\nu = K_{\max} + h\nu_0$
- $h\nu_0$**=2.50eV, given**

$$h\nu = h\frac{c}{\lambda} = 6.63 \times 10^{-34} Js\left(\frac{3 \times 10^{8} ms^{-1}}{3000 \times 10^{-10} m}\right) = 6.63 \times 10^{-19} J$$

$$= \frac{6.63 \times 10^{-19} J}{1.60 \times 10^{-19} \frac{J}{eV}} = 4.14 eV$$

$$K_{\max} = h\nu - h\nu_0 = 4.14\text{eV} - 2.50\text{eV} = 1.64\text{eV}$$

Further explanation on Quantum theory of light

Einstein Concept

- The energy in an electromagnetic (*e-m*) wave is carried in units called quanta or photons.
- Einstein proposed that light consists of particles called photons

Each photon carries energy $E = h\nu$ **, and**

Linear momentum $p = \frac{h}{\lambda}$

Using the relation $c = \nu\lambda$

$$p = h.\frac{\nu}{c} = \frac{h\nu}{c} = \frac{E}{c}$$

Where, p **= momentum;** ν **= frequency;**

λ **= wavelength;** c **=speed of light in vacuum;**

h **= Planck's constant and** E **= energy**

The Planck –Einstein Relations

- **The Planck–Einstein relations connect the particle-like properties of energy and momentum to wave-like properties of frequency and wave vector K.**

Using

$$E = hv; v = \frac{\omega}{2\pi}; and, \hbar = \frac{h}{2\pi}$$

For energy,

$$E = hv = h.\frac{\omega}{2\pi} = \frac{h}{2\pi}.\omega = \hbar\omega$$

For momentum,

$$p = \frac{h}{\lambda}; K = \frac{2\pi}{\lambda}; or \lambda = \frac{2\pi}{K}$$

Then,

$$p = \frac{h}{\lambda} = h.\frac{1}{\lambda} = h.\frac{K}{2\pi} = \frac{h}{2\pi}.K = \hbar K$$

Example

a. Calculate the energy in eV of a photon of wavelength $1\overset{0}{A}$.

b. What is the momentum of this photon?

($h = 6.63 \times 10^{-34} Js$, $c = 3 \times 10^{8} ms^{-1}$; $1\overset{0}{A} = 10^{-10} m$)

Solution:

a. $$E = \frac{hc}{\lambda} = \frac{6.63 \times 10^{-34} Js \times 3 \times 10^{8} ms^{-1}}{10^{-10} m} = 19.86 \times 10^{-16} J$$

$$E = \frac{19.86 \times 10^{-16} J}{1.6 \times 10^{-19} \frac{J}{eV}} = 1.24 \times 10^{4} eV$$

b. $$p = \frac{h}{\lambda} = \frac{6.63 \times 10^{-34} Js}{10^{-10} m} = 6.63 \times 10^{-24} Ns$$

Assignment 2

Calculate the maximum energy and speed of the emitted electrons when sodium is illuminated by a radiation of wavelength 150nm.
What is the least frequency of radiation (threshold frequency) for which electrons are emitted?
(*Take work function of sodium as 2.0 eV*).
Given that $m_e = 9.109 \times 10^{-31}$kg,
$m_n = 1.675 \times 10^{-27}$kg, and $m_p = 1.672 \times 10^{-27}$kg

ALL ASSIGNMENTS MUST BE HANDED IN VIA COURSE REPS
BEFORE THE NEXT CLASS

THANKS TO YOU ALL!

GOD BLESS YOU ALL IN JESUS NAME!